



AI Infrastructure Architecture Evolution

2015-2026

Silicon generations · Memory and interconnects
Conference disclosures · Systems and software

A generation-by-generation synthesis of conference disclosures and primary technical sources.
Evidence cutoff: 28 September 2026.
Complete English edition, paired with a full Simplified Chinese report.

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01 Scope and architectural synthesis

This report follows Microsoft-owned cloud silicon from the FPGA era to Cobalt CPUs, Maia accelerators and Azure Boost infrastructure offload, covering 2015-28 September 2026 with a 2014 Catapult baseline. AMD, Intel and NVIDIA systems in Azure provide deployment context; they are not relabelled Microsoft silicon. Xbox and client devices are outside the deep scope. Internal production, customer preview and public VM general availability are distinct milestones.

The continuity is explicit control over data movement. Catapult moved services into a reconfigurable network path; Maia exposes a hierarchy of scratchpads, DMA and synchronization; Boost removes infrastructure work from host CPUs. Cobalt supplies the general-purpose execution plane. These are complementary machines with different programming contracts. Their joint value depends on resource isolation and utilization across the service, rather than the largest accelerator FLOPS number.

02 The FPGA foundation: Catapult and Brainwave

The ISCA 2014 Catapult paper is a pre-period baseline for accelerating Bing ranking with distributed FPGAs. The MICRO 2016 configurable-cloud work shifts the emphasis toward a datacenter-wide reconfigurable fabric integrated with networking. Brainwave, described at ISCA 2018, maps real-time neural inference onto FPGAs. The progression matters: an accelerator becomes a service and network resource, not merely a peripheral attached to one application process.

Reconfigurability lets the operator move functions as protocols and models change, but consumes area and energy compared with dedicated logic. A later ASIC can harden stable operations while retaining programmable control. This is an architectural lineage, not evidence that Maia reuses a particular Catapult RTL block or that every Boost generation is an FPGA. Research prototypes, production infrastructure and public VM products must remain separate in the timeline.

03 Cobalt: general-purpose cloud compute

Cobalt 100 establishes Microsoft's own Arm server-CPU line with 128 Neoverse N2 cores. Its VMs became generally available in October 2024. The relevant comparison is a matched VM and service workload, including memory, storage and network limits; a processor core count alone does not describe the customer-visible machine.

Cobalt 200 moves to Neoverse CSS V3 and a chiplet implementation. Microsoft discloses 132 active cores, 3 MB L2 per core and 192 MB shared system cache. The June 2026 VM milestone is early-access preview, with VM sizes up to 128 vCPUs. The difference between 132 physical active cores and 128 customer vCPUs should be preserved. Larger caches target the cost of bringing data to the core; advertised service gains are not a universal IPC multiplier.

CPU generations · 1/3

n/d: not established; n/a: not applicable. Precision, scope and direction are shown in the cells; sources are mapped at the end.

Generation / reference SKU	Codename	GA	Status	Core µarch	Cores / threads
Cobalt 100	n/d	2024-10	Shipping / available	Neoverse N2	128 / 128
Cobalt 200	n/d	2026-06 preview	Early-access preview	Neoverse CSS V3	132 / 132

CPU generations · 2/3

n/d: not established; n/a: not applicable. Precision, scope and direction are shown in the cells; sources are mapped at the end.

Generation / reference SKU	Process per die	Dies / packaging	L2 per core	L3 / domain
Cobalt 100	n/d	n/d	n/d	n/d
Cobalt 200	3 nm	Chiplets; count n/d	3 MB	192 MB / SoC

CPU generations · 3/3

n/d: not established; n/a: not applicable. Precision, scope and direction are shown in the cells; sources are mapped at the end.

Generation / reference SKU	Memory: channels; rate; GB/s	Sockets / links	PCIe / CXL	TDP	Vector / matrix ISA
Cobalt 100	DDR5; channels/rate n/d	n/d	n/d	n/d	Armv9
Cobalt 200	n/d	n/d	n/d	n/d	Armv9

04 Maia 100: building a software-defined machine

Hot Chips 2024 supplies the implementation picture missing from the original launch: an approximately 820 mm² N5 die, CoWoS-S packaging, 64 GB HBM2e at 1.8 TB/s and roughly 500 MB of local SRAM. Sixteen clusters each contain four tiles, with tensor, vector, data-movement and control functions. The slide specification separates a 700 W design target from 500 W provisioned power. Those are not interchangeable operating points.

The compiler orchestrates asynchronous engines using semaphores. Triton offers a higher-level route, while the Maia API exposes explicit placement and scheduling. This trades automatic cache management for control over reuse and overlap. A good schedule can fuse elementwise work with matrix operations and communication; a poor schedule can stall a large tensor engine despite abundant peak compute. Six- and nine-bit tensor formats in the slides must not be renamed FP4 and FP8.

05 Maia 200: dataflow extends through the network

The 2026 technical paper makes Maia 200 a software-defined dataflow system rather than a matrix engine with an attached NIC. A monolithic 3 nm compute die uses six HBM3e stacks, 216 GB capacity and 7 TB/s bandwidth. Four clusters contain nine or ten tiles each. Separate data and control networks, specialized SRAM, DMA and synchronization engines let software describe when data is ready and where it must move. The published peak is 10,145 TFLOP/s FP4 and 5,072 TFLOP/s FP8 within a 750 W TDP.

Twenty-eight integrated 400 Gb/s controllers supply 1.4 TB/s in each direction. Twenty ports use fixed links and eight feed four switched planes. The described deployment design point is 6,144 chips with oversubscription in the external fabric; it is not a fully connected 6,144-chip memory domain. ATLv2 provides transport, encryption, selective retransmission and congestion control. Remote SRAM access uses explicit send/receive and synchronization, not CPU-style transparent cache coherence.

Architectural interpretation: integrating communication into the dataflow contract can reduce launch and host-interface overhead, but transfers scheduling responsibility into the compiler and runtime. The practical test is whether compute, memory traffic and collectives overlap for the actual model shape. More low-precision arithmetic increases the reuse required to remain compute-bound; it does not remove the HBM or network roofline.

Accelerator generations · 1/4

n/d: not established; n/a: not applicable. Precision, scope and direction are shown in the cells; sources are mapped at the end.

Product	Architecture	GA / milestone	Status	Process	Dies / packaging
Maia 100	Software-defined dataflow	2023 launch; HC2024	Historical shipping	TSMC N5	1 logic die; CoWoS-S
Maia 200	SDLA	2026 deployment	Shipping / available	3 nm	1 logic die; 75×75 mm package

Accelerator generations · 2/4

n/d: not established; n/a: not applicable. Precision, scope and direction are shown in the cells; sources are mapped at the end.

Product	Compute units	Dense BF16	Dense FP8	Dense FP6 / FP4	FP64 vector / matrix
Maia 100	64 tiles / 16 clusters	800 TFLOP/s	n/a; 9-bit: 1.5 POPS	6-bit: 3 POPS; FP4 n/d	n/d
Maia 200	4 clusters; 9-10 tiles/cluster	n/d	5072 TFLOP/s	FP4: 10145 TFLOP/s	n/d

Accelerator generations · 3/4

n/d: not established; n/a: not applicable. Precision, scope and direction are shown in the cells; sources are mapped at the end.

Product	Memory: type; stacks; GB; TB/s	LLC / SRAM
Maia 100	HBM2e; stacks n/d; 64 GB; 1.8 TB/s	~500 MB
Maia 200	HBM3e; 6; 216 GB; 7 TB/s	272 MB

Accelerator generations · 4/4

n/d: not established; n/a: not applicable. Precision, scope and direction are shown in the cells; sources are mapped at the end.

Product	Scale-up: links; GB/s per direction; domain	Host link	Power / cooling	Form factor
Maia 100	12×400G; 600 GB/s one-way; domain n/d	PCIe5 ×8 (spec table)	700 W design / 500 W provision	Azure module
Maia 200	28×400G; 1400 GB/s one-way; topology-specific	PCIe6 ×8	750 W; liquid	Azure module

The Maia 100 slide block diagram contains an older PCIe label, while its specification table states PCIe5 ×8; this report uses the specification table and preserves the discrepancy. Maia 200’s paper uses both TB/s and TiB/s wording for memory bandwidth; the main specification and vendor architecture disclosure use 7 TB/s, the basis for calculations here. No measured application efficiency is inferred from TDP and peak FLOPS.

06 Azure Boost, AI NICs and security

Azure Boost is an infrastructure system; the Azure Boost DPU is a silicon component in that evolution. The 2024 DPU announcement and the 2026 general availability of a newer Boost generation describe hardening more storage and networking functions into dedicated hardware. MANA supplies the guest-facing network interface. Up to 400 Gb/s is a supported platform/VM ceiling, not evidence that every VM receives that rate. Cerberus anchors the hardware trust boundary.

Infrastructure and integrated NICs · 1/4

n/d: not established; n/a: not applicable. Precision, scope and direction are shown in the cells; sources are mapped at the end.

Product	GA / milestone	Status	Class
Azure Boost / DPU	2024 announcement; 2026 system GA	Shipping / available	Infrastructure offload
Maia 200 ANC	2026	Shipping / available	Integrated AI NIC

Infrastructure and integrated NICs · 2/4

n/d: not established; n/a: not applicable. Precision, scope and direction are shown in the cells; sources are mapped at the end.

Product	Ports × speed	SerDes	Host interface	Programmable engine
Azure Boost / DPU	Up to 400 Gb/s platform	n/d	n/d	Dedicated offload logic
Maia 200 ANC	28×400 Gb/s	n/d	On-chip NoC	DMA / ATLv2

Infrastructure and integrated NICs · 3/4

n/d: not established; n/a: not applicable. Precision, scope and direction are shown in the cells; sources are mapped at the end.

Product	Embedded CPU	On-card memory	Transport / RDMA	Congestion control
Azure Boost / DPU	n/d	n/d	MANA / RDMA	n/d
Maia 200 ANC	n/a	Shared chip memories	ATLv2 / Ethernet	Window-based; ECMP / entropy

Infrastructure and integrated NICs · 4/4

n/d: not established; n/a: not applicable. Precision, scope and direction are shown in the cells; sources are mapped at the end.

Product	Key offloads	Power
Azure Boost / DPU	Network; storage; virtualization; trust	n/d
Maia 200 ANC	Remote memory; encryption; retransmit	Included in SoC TDP

Interconnect hierarchy · 1/2

n/d: not established; n/a: not applicable. Precision, scope and direction are shown in the cells; sources are mapped at the end.

Interconnect	Year / generation	Lane rate	Lanes / link	GB/s per link per direction
Maia100 Ethernet	2024	400 Gb/s port	n/d	50
Maia200 fixed links	2026	400 Gb/s port	n/d	50
Maia200 switched links	2026	400 Gb/s port	n/d	50

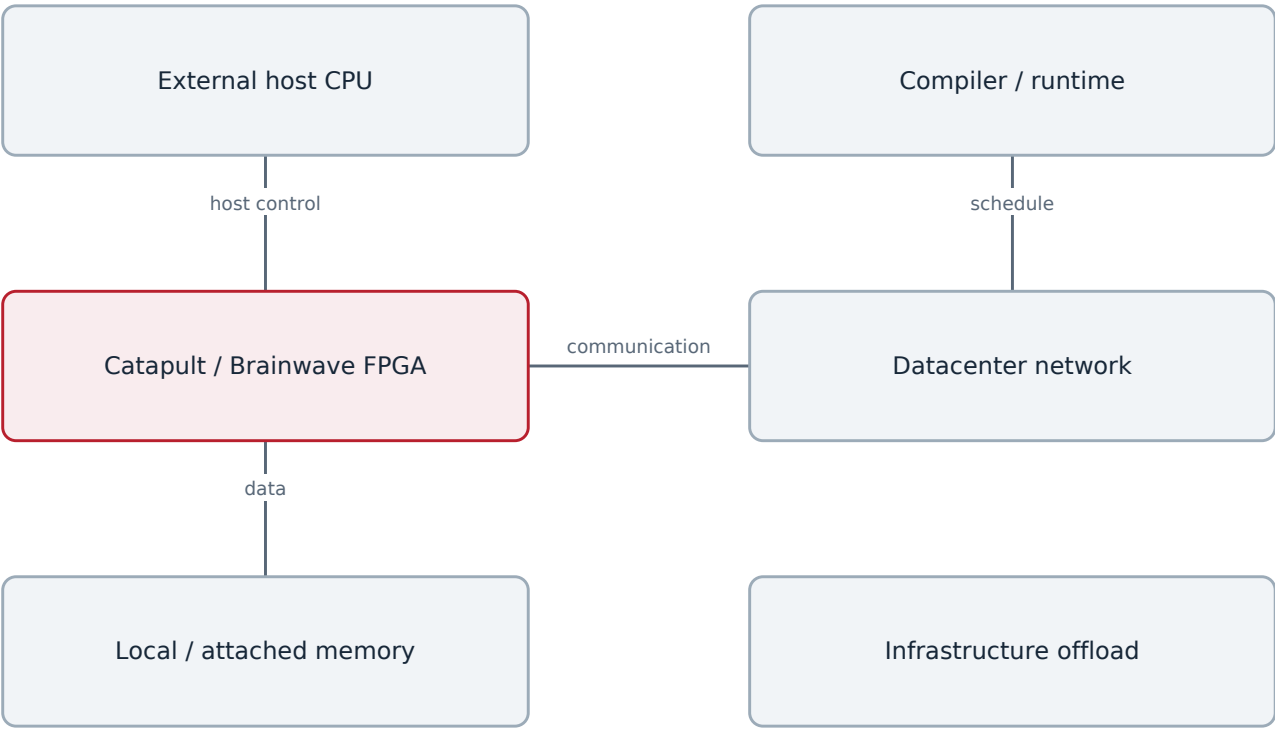
Interconnect hierarchy · 2/2

n/d: not established; n/a: not applicable. Precision, scope and direction are shown in the cells; sources are mapped at the end.

Interconnect	Links / device	Topology	Domain limit	Coherence / semantics
Maia100 Ethernet	12	Direct + switched	n/d	Explicit transfer
Maia200 fixed links	20	Tray fixed links	4 chips/tray	Explicit SDLA transfer
Maia200 switched links	8	4 planes; 2 tiers	6144 design point	Explicit transfer; not coherent

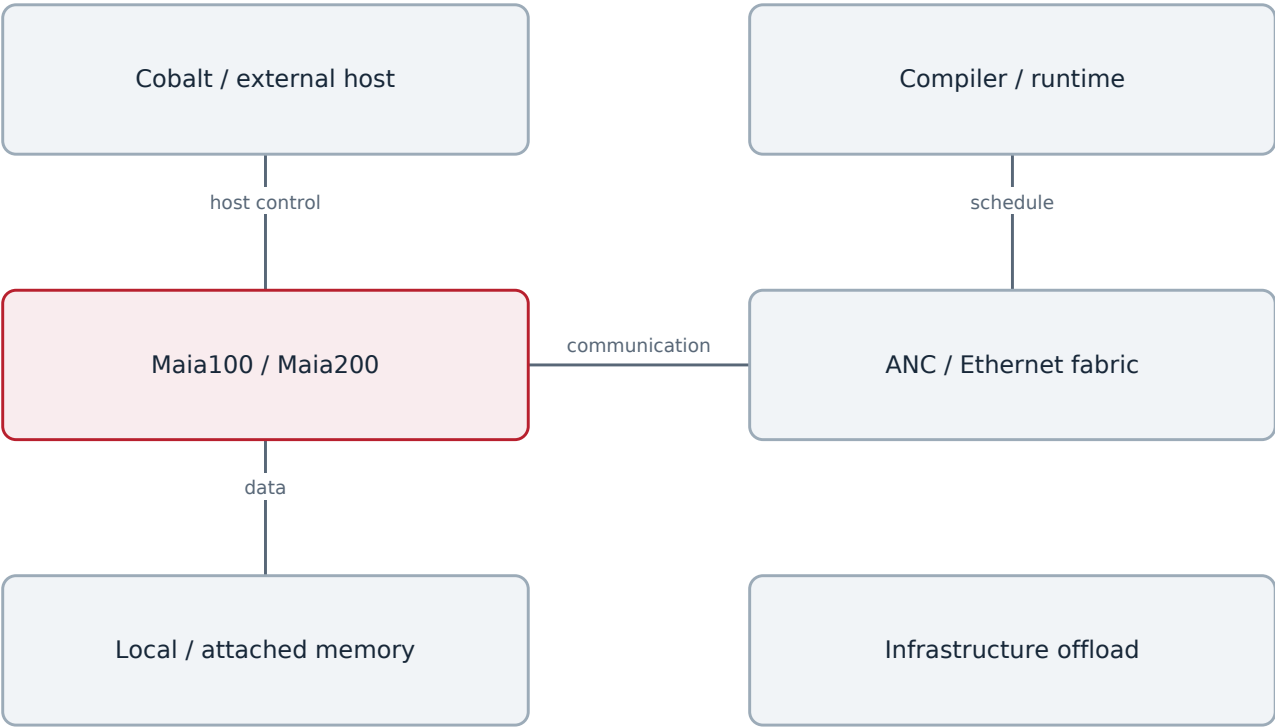
07 Systems and programming stack

2015-2018 system roles



Functional relationships; not a claim that every Maia host is Cobalt.

2024-2026 system roles



Functional relationships; not a claim that every Maia host is Cobalt.

System scope · 1/2

n/d: not established; n/a: not applicable. Precision, scope and direction are shown in the cells; sources are mapped at the end.

Platform	Year / milestone	CPU / accelerator count	Scale-up domain
Maia100 Azure	2023-2024	n/d	Topology-specific
Maia200 deployment design	2026	4 chips/tray; up to 6144	Fixed links + switched planes
Cobalt200 VMs	2026 preview	Up to 128 vCPUs	n/a

System scope · 2/2

n/d: not established; n/a: not applicable. Precision, scope and direction are shown in the cells; sources are mapped at the end.

Platform	Scale-out per accelerator	Rack power	Cooling
Maia100 Azure	n/d	n/d	Liquid
Maia200 deployment design	8×400G = 400 GB/s nominal	n/d	Liquid
Cobalt200 VMs	VM-specific	n/d	n/d

Software contracts

n/d: not established; n/a: not applicable. Precision, scope and direction are shown in the cells; sources are mapped at the end.

Layer	Component	Function	Boundary
Host	Arm Linux / Azure VMs	General-purpose services	VM limits differ from SoC
Accelerator	PyTorch / Triton / Maia API	Graph, kernels, placement	Compiler-managed scratchpads
Communication	Maia collectives / ATLv2	Collectives and transport	Explicit remote data movement
Infrastructure	MANA / Boost / Cerberus	Networking, storage, trust	Platform and guest drivers

08 Normalized balance and trends

Derived metrics

n/d: not established; n/a: not applicable. Precision, scope and direction are shown in the cells; sources are mapped at the end.

Metric	Calculation	Meaning
Cobalt200 L3/core	$192/132 = 1.455 \text{ MB}$	Shared capacity / active core
Maia100 HBM/BF16	$1.8e12/800e12 = 0.00225 \text{ B/FLOP}$	Dense peak balance
Maia100 capacity/BF16	$64e9/800e12 = 8.0e-5 \text{ B}\cdot\text{s/FLOP}$	Capacity normalized by peak
Maia200 HBM/FP8	$7e12/5072e12 = 0.00138 \text{ B/FLOP}$	Different format from row above
Maia200 capacity/FP8	$216e9/5072e12 = 4.26e-5 \text{ B}\cdot\text{s/FLOP}$	Per chip; decimal bytes
Maia200 network/HBM	$1.4/7 = 0.20$	All ports one-way; not all external
Maia200 FP8/TDP	$5072/750 = 6.763 \text{ TFLOP/s/W}$	Peak/TDP proxy; not measured efficiency
Cobalt DDR bandwidth/core	n/d	Matched memory-controller data absent

The Maia transition increases HBM capacity by 3.375× and bandwidth by 3.89×, while total integrated network bandwidth grows 2.33×. It also changes numerical formats and the on-chip memory organization, so a single compute-growth ratio would hide important differences. For deployment decisions, measure latency under realistic batching, scratchpad occupancy, communication overlap and failure recovery. These expose whether the custom architecture improves a service rather than only a kernel.

09 Conference-to-product index

Primary disclosure map · 1/2

n/d: not established; n/a: not applicable. Precision, scope and direction are shown in the cells; sources are mapped at the end.

Venue	Year	Paper / talk / disclosure
ISCA	2014	Catapult
MICRO	2016	A Configurable Cloud
ISCA	2018	Brainwave
Hot Chips	2024	Inside Maia100
Hot Chips	2026	Maia200
Ignite / Build	2023-2026	Cobalt / Maia / Boost

Primary disclosure map · 2/2

n/d: not established; n/a: not applicable. Precision, scope and direction are shown in the cells; sources are mapped at the end.

Venue	Presenter / authors	Product	Evidence type	Source
ISCA	Putnam et al.	FPGA infrastructure	Pre-period research baseline	C04
MICRO	Microsoft Research	Catapult	Architecture / deployment	P03
ISCA	Microsoft Research	FPGA inference	Architecture paper	C05
Hot Chips	Sherry Xu; Chandru Ramakrishnan	Maia100	Full technical slides	C06
Hot Chips	Microsoft	Maia200	Program plus companion technical paper	C03 C01
Ignite / Build	Microsoft	Cloud portfolio	Launch and availability	E03 E02 E05 E06

ISCA and MICRO establish the FPGA lineage; Hot Chips supplies the Maia block and implementation views. ISSCC2026 paper 17.4 establishes an additional Maia implementation disclosure; the official program does not specify a generation suffix. This edition does not establish a Cobalt/Maia product-specific HPCA or ASPLOS paper. A conference program proves a presentation exists; detailed specifications here come from the slides, technical paper and product documentation. The main remaining gaps are complete Cobalt memory/PCIe specifications, matched Maia operating points and SKU-level Boost implementation details.

ISSCC implementation disclosure · 1/2

n/d: not established; n/a: not applicable. Precision, scope and direction are shown in the cells; sources are mapped at the end.

Venue	Year	Paper / talk / disclosure
ISSCC	2026	17.4 MAIA: A Reticle-Scale AI Accelerator

ISSCC implementation disclosure · 2/2

n/d: not established; n/a: not applicable. Precision, scope and direction are shown in the cells; sources are mapped at the end.

Venue	Presenter / authors	Product	Evidence type	Source
ISSCC	S. Xu et al.	Maia; generation suffix not in program	Official program / published paper record; full digest not accessed	C07

The ISSCC record establishes a circuit/implementation disclosure alongside the Hot Chips architectural chain. Because the full digest was not accessible in this review, its title is not used to assign undisclosed clocking, voltage-regulator or SRAM-circuit details to Maia100 or Maia200. The full Hot Chips slides and the Maia200 technical paper remain the quantitative implementation sources.

10 Naming crosswalk

Products and roles

n/d: not established; n/a: not applicable. Precision, scope and direction are shown in the cells; sources are mapped at the end.

Name	Role	Do not conflate with
Cobalt	General-purpose host CPU	Maia
Maia / ANC	AI accelerator / integrated network controller	Boost DPU
Azure Boost	Infrastructure system including specialized hardware	One fixed chip generation
MANA	Microsoft Azure Network Adapter	Maia backend fabric

Source map

- 01 Scope and architectural synthesis - C01, C04, C06, E02, E03, P03, P04
- 02 The FPGA foundation: Catapult and Brainwave - C04, C05, E06, P03
- 03 Cobalt: general-purpose cloud compute - E02, E07, E08, P01
- 04 Maia 100: building a software-defined machine - C06
- 05 Maia 200: dataflow extends through the network - C01, C06, P02
- 06 Azure Boost, AI NICs and security - C01, C06, E05, E06, P04
- 07 Systems and programming stack - C01, C06, E02, E06, E07, P04
- 08 Normalized balance and trends - C01, C06, P01
- 09 Conference-to-product index - C01, C03, C04, C05, C06, C07, E02, E03, E05, E06, P01, P03, P04
- 10 Naming crosswalk - C01, E02, E06, P04

References

Product and technical documentation

[P01] Cobalt200 architecture announcement

<https://techcommunity.microsoft.com/blog/AzureInfrastructureBlog/announcing-cobalt-200-azure%E2%80%99s-next-cloud-native-cpu/4469807>

Indexed record or searchable official page; product details cross-checked with the associated technical document.

[P02] Maia200 architecture deep dive

<https://techcommunity.microsoft.com/blog/azureinfrastructureblog/deep-dive-into-the-maia-200-architecture/4489312/>

Indexed record or searchable official page; product details cross-checked with the associated technical document.

[P03] Catapult publications and MICRO2016

<https://www.microsoft.com/en-us/research/project/project-catapult/publications/>

Official indexed record or related announcement; full document download unavailable.

[P04] Azure Boost overview

<https://learn.microsoft.com/en-us/azure/azure-boost/overview>

Primary source; accessed by 2026-09-28

Conference records and implementation disclosures

[C01] Maia200: A Software Defined Dataflow System, 2026

<https://arxiv.org/pdf/2608.24664>

Primary source; accessed by 2026-09-28

[C03] Hot Chips2026 program, Maia200

<https://hc2026.hotchips.org/program/conference/>

Official indexed record or related announcement; full document download unavailable.

[C04] Catapult ISCA2014

https://www.microsoft.com/en-us/research/wp-content/uploads/2016/02/Catapult_ISCA_2014.pdf

Official indexed record or related announcement; full document download unavailable.

[C05] Brainwave ISCA2018

<https://www.microsoft.com/en-us/research/uploads/prod/2018/06/ISCA18-Brainwave-CameraReady.pdf>

Official indexed record or related announcement; full document download unavailable.

[C06] Inside Maia100, Hot Chips2024 slides

https://hc2024.hotchips.org/assets/program/conference/day2/81_HC2024.Microsoft.Xu.Ramakrishnan.final.v2.pdf

Primary source; accessed by 2026-09-28

[C07] ISSCC2026 advance program; 17.4 MAIA: A Reticle-Scale AI Accelerator

<https://submissions.mirasmart.com/ISSCC2026/PDF/ISSCC2026AdvanceProgram.pdf>

Official conference program or press material; not the full paper.

Launches, ecosystem and deployment

[E02] Cobalt200 early access preview, June2026

<https://azure.microsoft.com/en-us/blog/new-azure-cobalt-200-vms-deliver-50-performance-improvement-fully-optimized-for-modern-agentic-ai-workloads/>

Indexed record or searchable official page; product details cross-checked with the associated technical document.

[E03] Ignite2023 Cobalt Maia Boost

<https://news.microsoft.com/source/features/ai/in-house-chips-silicon-to-service-to-meet-ai-demand/>

Official indexed record or related announcement; full document download unavailable.

[E05] Azure Boost DPU, Ignite2024

<https://techcommunity.microsoft.com/blog/azureinfrastructureblog/enhancing-infrastructure-efficiency-with-azure-boost-dpu/4298901>

Indexed record or searchable official page; product details cross-checked with the associated technical document.

[E06] Next generation Azure Boost GA, 2026

<https://techcommunity.microsoft.com/blog/azurecompute/announcing-the-general-availability-of-the-next-generation-of-azure-boost/4519136>

Indexed record or searchable official page; product details cross-checked with the associated technical document.

[E07] Cobalt100 VM general availability, October2024

<https://azure.microsoft.com/en-us/blog/azure-cobalt-100-based-virtual-machines-are-now-generally-available/>

Indexed record or searchable official page; product details cross-checked with the associated technical document.

[E08] Arm earnings disclosure: Cobalt N2 and V3

<https://investors.arm.com/static-files/941fd4cb-0027-4e81-9541-66dcd954a14a>

Primary source; accessed by 2026-09-28